

Loan Default Detection Using Machine Learning

1. Introduction

Loan default prediction is an important task in the banking and finance sector. Financial institutions need reliable methods to determine whether a customer is likely to repay a loan. Incorrect lending decisions can result in financial losses and increased risk. Machine learning techniques can assist banks by analyzing historical loan application data and predicting loan approval or default outcomes.

The objective of this project is to build and compare machine learning models capable of predicting loan status based on applicant information such as income, credit history, education, and property area.

2. Dataset Description

The dataset used in this project was obtained from Kaggle's Loan Prediction Dataset. It contains 614 records and 13 attributes related to loan applicants.

Important Features

- Gender
- Married
- Dependents
- Education
- Self_Employed
- ApplicantIncome
- CoapplicantIncome
- LoanAmount
- Loan_Amount_Term
- Credit_History
- Property_Area
- Loan_Status (Target Variable)

Target Variable

Loan_Status:

- Y = Loan Approved
- N = Loan Not Approved / Default Risk

	Loan_ID	Gender	Married	Dependents	Education	Self_Employed	\
0	LP001002	Male	No	0	Graduate	No	
1	LP001003	Male	Yes	1	Graduate	No	
2	LP001005	Male	Yes	0	Graduate	Yes	
3	LP001006	Male	Yes	0	Not Graduate	No	
4	LP001008	Male	No	0	Graduate	No	

	ApplicantIncome	CoapplicantIncome	LoanAmount	Loan_Amount_Term	\
0	5849	0.0	NaN	360.0	
1	4583	1508.0	128.0	360.0	
2	3000	0.0	66.0	360.0	
3	2583	2358.0	120.0	360.0	
4	6000	0.0	141.0	360.0	

	Credit_History	Property_Area	Loan_Status
0	1.0	Urban	Y
1	1.0	Rural	N
2	1.0	Urban	Y
3	1.0	Urban	Y
4	1.0	Urban	Y

```

<class 'pandas.DataFrame'>
RangeIndex: 614 entries, 0 to 613
Data columns (total 13 columns):
#   Column                Non-Null Count  Dtype
---  -
...
25%    1.000000
50%    1.000000
75%    1.000000
max     1.000000

```

3. Methodology

3.1 Data Preprocessing

The dataset was cleaned and prepared before model training. Missing values in numerical columns were replaced using median values, while categorical columns were filled using mode values.

```
Loan_ID      0
Gender       13
Married       3
Dependents   15
Education    0
Self_Employed 32
ApplicantIncome 0
CoapplicantIncome 0
LoanAmount   22
Loan_Amount_Term 14
Credit_History 50
Property_Area 0
Loan_Status  0
dtype: int64
```

Categorical features were converted into numerical values using Label Encoding. Data preprocessing ensured that the dataset was suitable for machine learning algorithms.

3.2 Feature Engineering

Two new features were created:

- $\text{TotalIncome} = \text{ApplicantIncome} + \text{CoapplicantIncome}$
- $\text{LoanIncomeRatio} = \text{LoanAmount} / \text{TotalIncome}$

These features were added to improve the predictive power of the models.

3.3 Data Splitting

The dataset was divided into:

- Training Set: 80%
- Testing Set: 20%

This allowed the models to be trained and evaluated on separate data.

3.4 Machine Learning Models

Two classification algorithms were selected:

Decision Tree Classifier

A tree-based supervised learning algorithm that makes decisions using a sequence of feature-based rules.

K-Nearest Neighbors (KNN)

A distance-based algorithm that classifies data points based on the majority class of their nearest neighbors.

4. Evaluation Metrics

The following metrics were used to evaluate model performance:

- Accuracy
- Precision
- Recall
- F1 Score
- Confusion Matrix

These metrics provide a comprehensive understanding of model effectiveness.

5. Results

Model Comparison

Metric	Decision Tree	KNN
Accuracy	77.24%	75.61%
Precision	76.00%	75.00%
Recall	95.00%	93.75%
F1 Score	84.44%	83.33%

Analysis

The Decision Tree model achieved the highest performance across all evaluation metrics.

- Decision Tree Accuracy: 77.24%
- KNN Accuracy: 75.61%

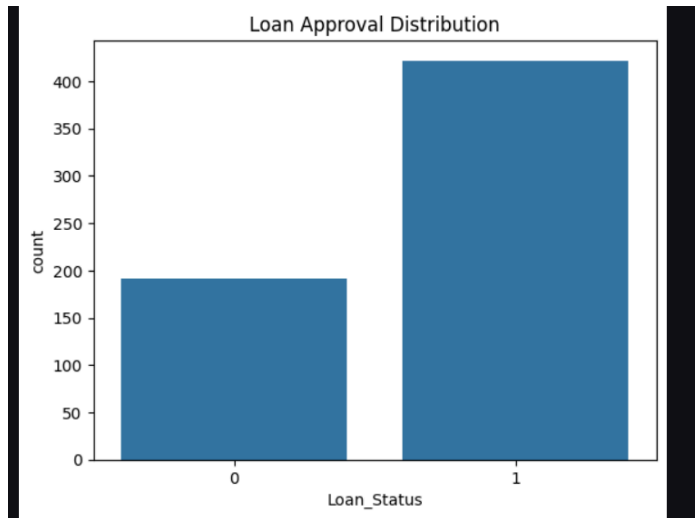
Decision Tree also produced slightly higher Precision, Recall, and F1 Score values compared to KNN.

Confusion matrix visualizations further confirmed that the Decision Tree model produced more accurate classifications.

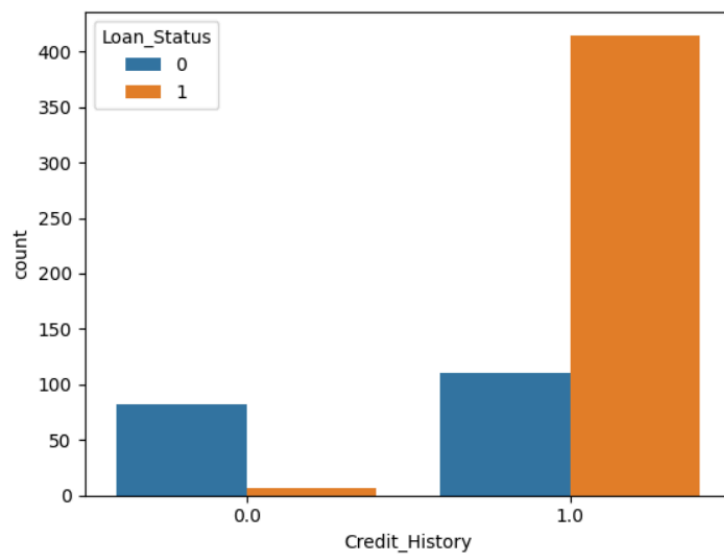
6. Visualizations

The following visualizations were generated during the project:

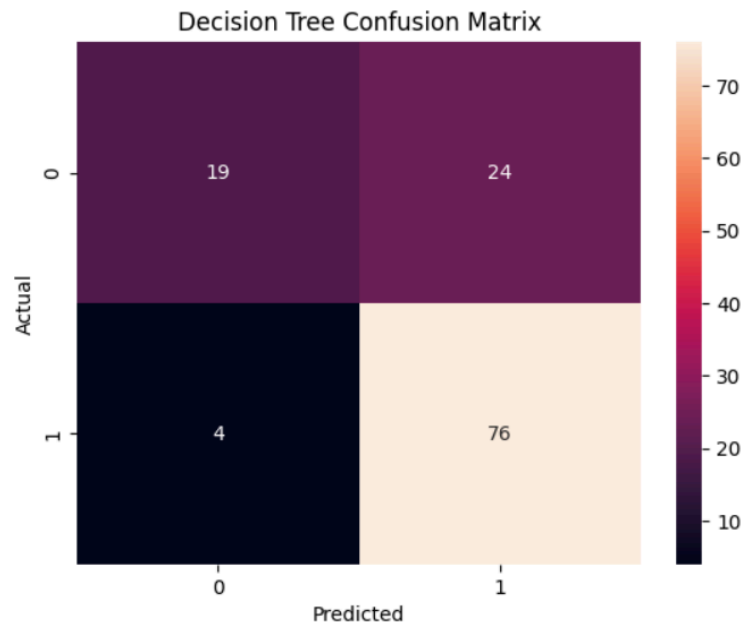
- Loan Approval Distribution



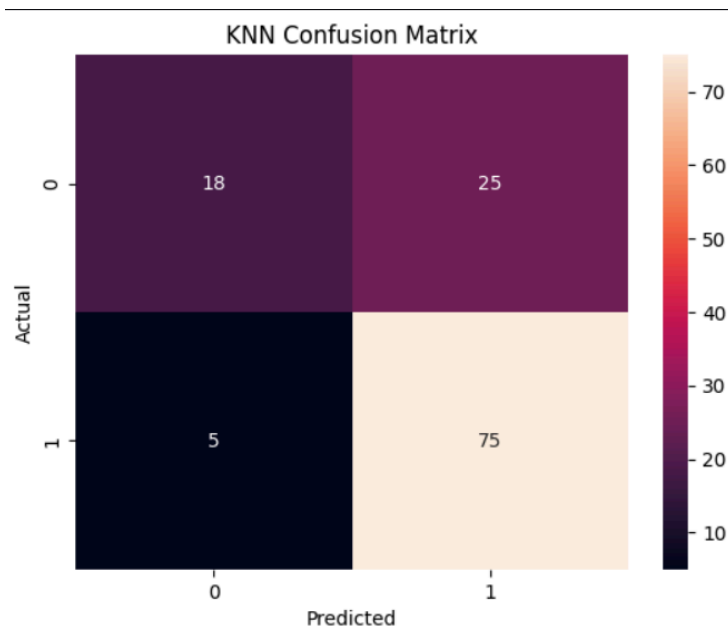
- Credit History vs Loan Status



- Decision Tree Confusion Matrix



- KNN Confusion Matrix



These visualizations helped identify patterns within the dataset and compare model performance.

7. Conclusion

This project successfully implemented a complete machine learning workflow for loan default detection. The dataset was preprocessed, new features were engineered, and two machine learning models were trained and evaluated.

Based on the evaluation results, the Decision Tree classifier outperformed the K-Nearest Neighbors model with an accuracy of 77.24% and an F1 Score of 84.44%. Therefore, Decision Tree was selected as the preferred model for loan default prediction.

The project demonstrates how machine learning can assist financial institutions in making faster and more informed lending decisions while reducing the risk associated with loan defaults.

References

1. Kaggle Loan Prediction Dataset
2. Scikit-learn Documentation
3. Pandas Documentation
4. NumPy Documentation